2.1
INPUT DEVICES

THESE ARE THE DEVICES THAT ENTER DATA INTO A COMPUTER TO PERFORM TASKS

NUMERIC KEYPADS

Used for entering numbers as **input**. They can be found on ATMs, Mobile phones, at markets to submit payment or on normal keyboards.They can be faster than normal keyboards when entering **numeric** data, and they are **smaller** and easier to carry around, but the buttons are also smaller which can produce difficulty.

KEY BOARDS

The most common and useful method for data entry. Keyboards are connected to the device and are used for entering fixed values for the computer to process and store or to give commands, it consists of the alphabetical letters of a specific language along with numbers and all the characters needed to keep grammar in line.

Most common type is **QWERTY**. It gets its name from the arrangement of letters on the top line of keys. There are also **AZERTY** and **QWERTZ** .

KEY BOARDS

These devices are

- the easiest and
- most common way of data entry, because of its convenience to most people,
- but it could be leading to injuring when used in a frequent and continuous way such as Repetitive Strain Injury (RSI).
- Errors can be easily made,
- also each country are required to have their own keyboards for different languages.
- Useless for artwork and diagram creation

Pointing Device

1 MOUSE

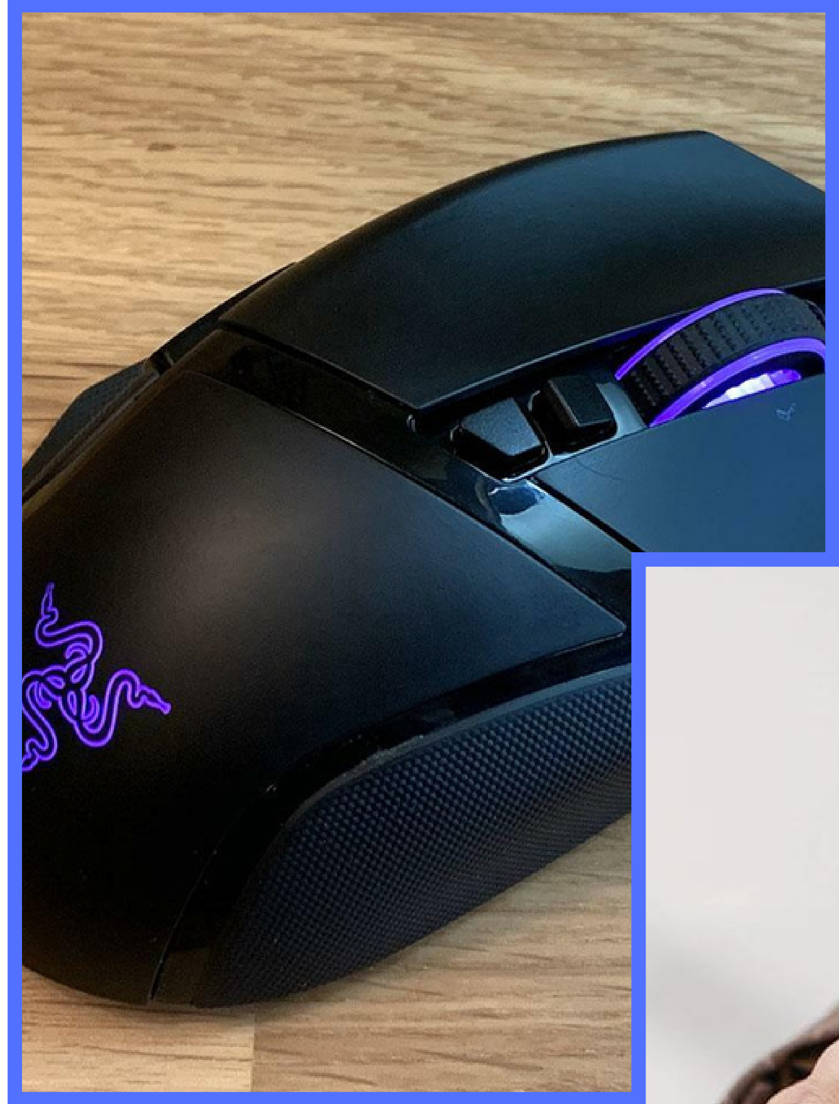
User controls the position of a pointer on the screen by moving the mouse, consists of two buttons and a scroll button, could be an **optical** mouse using lasers for movement or could track movement by having a ball underneath (**mechanical** mouse).

2 TOUCH PAD

A pointing device that uses hand movement as a way to position the pointer on the screen, also consists of two buttons, these are found on **laptops** which contribute to its portability, and **don't need flat surfaces** to operate like the mouse does.

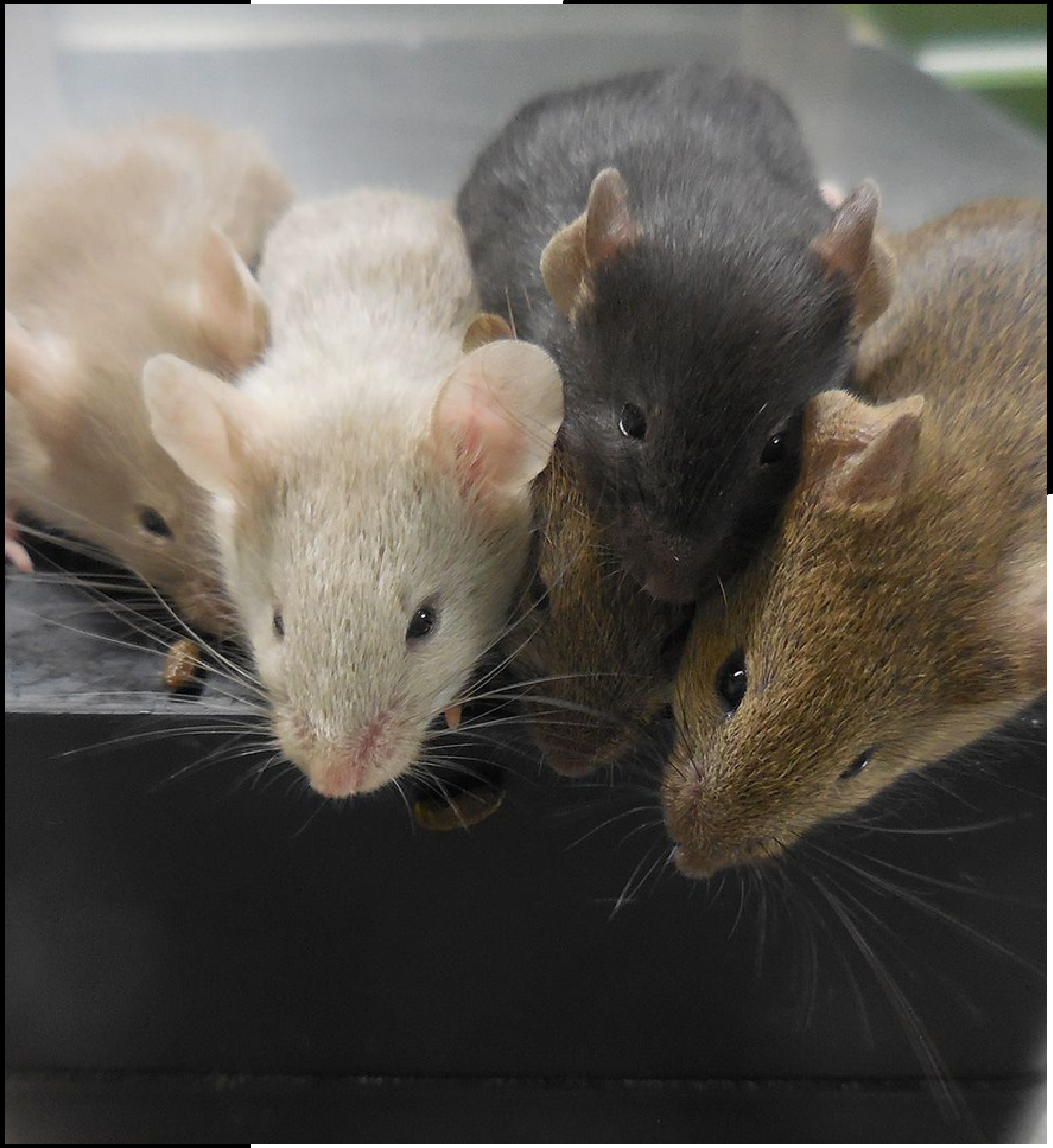
3 TRACKERBALL

Similar to mouse, but the ball here is on the top or sides of the device, which you would **use the ball to control** the pointer's movement. it also has buttons to select, but it does **not** need to physically move. It is however **difficult to use** for some applications.



MICE

A small rodent that typically has a pointed snout, relatively large ears and eyes, and a long tail.



MICE

Advantages:

- Fast input method
- Intuitive to use, you just point

Disadvantages:

- Easily damaged
- Disabled people may have difficulty
- Overuse can cause (RSI)
- Need flat surface to operate

REMOTE CONTROL

Uses **infrared signals** to operate other devices, each signal contains a code. There are codes for each button. Signals give commands to devices as input to control functions on another device from distance. Main **drawback** is that its signal can be blocked by objects.

DRIVING WHEELS

An input device that connects to a computer or a gaming console, that simulates driving a vehicle, along with pedals to accelerate and brake. It also has buttons to use as a normal controlling device. It is an input device that feels more **natural** than simple buttons to mimic controlling a vehicle, it can be **expensive** when strong and natural feedback are included tho.

JOY STICKS

An input device similar to mouse and trackerball, it consists of a stick and a few buttons, you control the pointer on the screen through the movement of the stick which you grip, and press buttons to give commands. It is used in **video games** mostly, but also in **flight simulators**, to mimic flying an airplane. It is **easier** than a keyboard to navigate the screen but more **difficult** to control the pointer compared to a mouse.

TOUCH SCREENS (AS AN INPUT)

These can be found on PCs, laptops, smartphones and tablets. A user can interact with it with their fingers. Two type exist:

- **Resistive**: Sensitive to pressure from finger or other objects. The touch point is detected due to 2 **metallic layers touching**. These require more pressure to use.

- **Capacitive**: Sensitive to static electricity on an exposed finger, its touch point detected by **sensors** at 4 corners of screen. These are more sensitive and responsive compared to Resistive.

TOUCH SCREENS (AS AN INPUT)

Advantages:

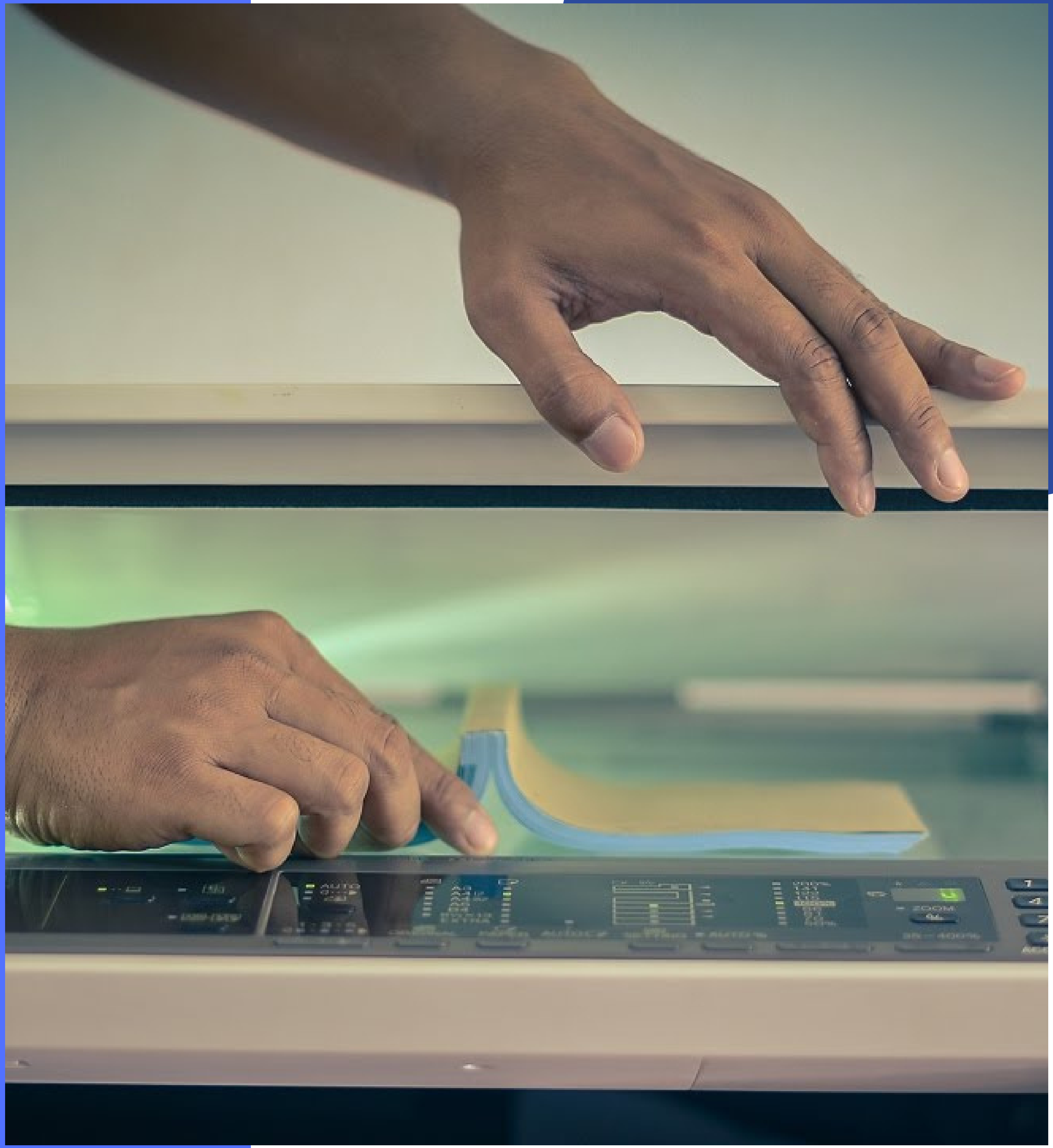
- Intuitive and easy to use
- Space-saving, input and output both in 1 device
- Cost-effective, less staff required because people can serve themselves

Disadvantages:

- Not suitable for large amounts of data
- Not accurate for small area selection
- Can be placed too high for disabled people
- Hygiene problems

SCANNERS

Used to enter information from **hard copy** into a computer. The hard copy document is scanned by a light source and produces a computer-readable image. These devices are used to **digitalize physical documents**, to protect the originals and make this of use or as means of archiving.



SCANNERS

Advantages:

- Very accurate and high quality images can be produced
- Hard-copies can be changed into a form that can be edited and stored on a computer
- Any document, or photo that is scanned, can then be used on electronic documents

Disadvantages:

- Images can still lose their original quality and be distorted
- Scanned images can take up a lot of storage space

Digital Camera

These largely replaced their older counter part, film-based cameras. The pictures here are stored on a **memory card (SSD)** and can be viewed within the camera screen or transferred to a computer by slotting its memory card into the computer, a **USB** port, or with **Wi-Fi** or **Bluetooth**. Cameras still do take **better** photos than smartphones, because of the dedicated software and use of expensive lenses.

Digital Camera

Advantages of digital cameras over the traditional film cameras include:

- Images can be seen and immediately deleted if not good enough
- Digital images can be edited more readily and easier
- The image can be uploaded onto a computer and be used in electronic documents

Disadvantages:

- Its memory card may become full and not save images
- Memory card's data can also get corrupted
- Cameras battery can run out and go off

Video cameras and webcams

Video cameras are used to record moving pictures, that are stored electronically. What is recorded is stored on a memory card, or directly feeding the video into the computer. Once it’s stored, it can be edited and used.

Video cameras can be used for leisure, producing commercials, or security purposes using CCTV .

Webcams are a special category of video camera that have no storage capacity but are **connected directly** to the computer. Laptops may have a built in webcam, or it can be a separate hardware plugged into a computer.

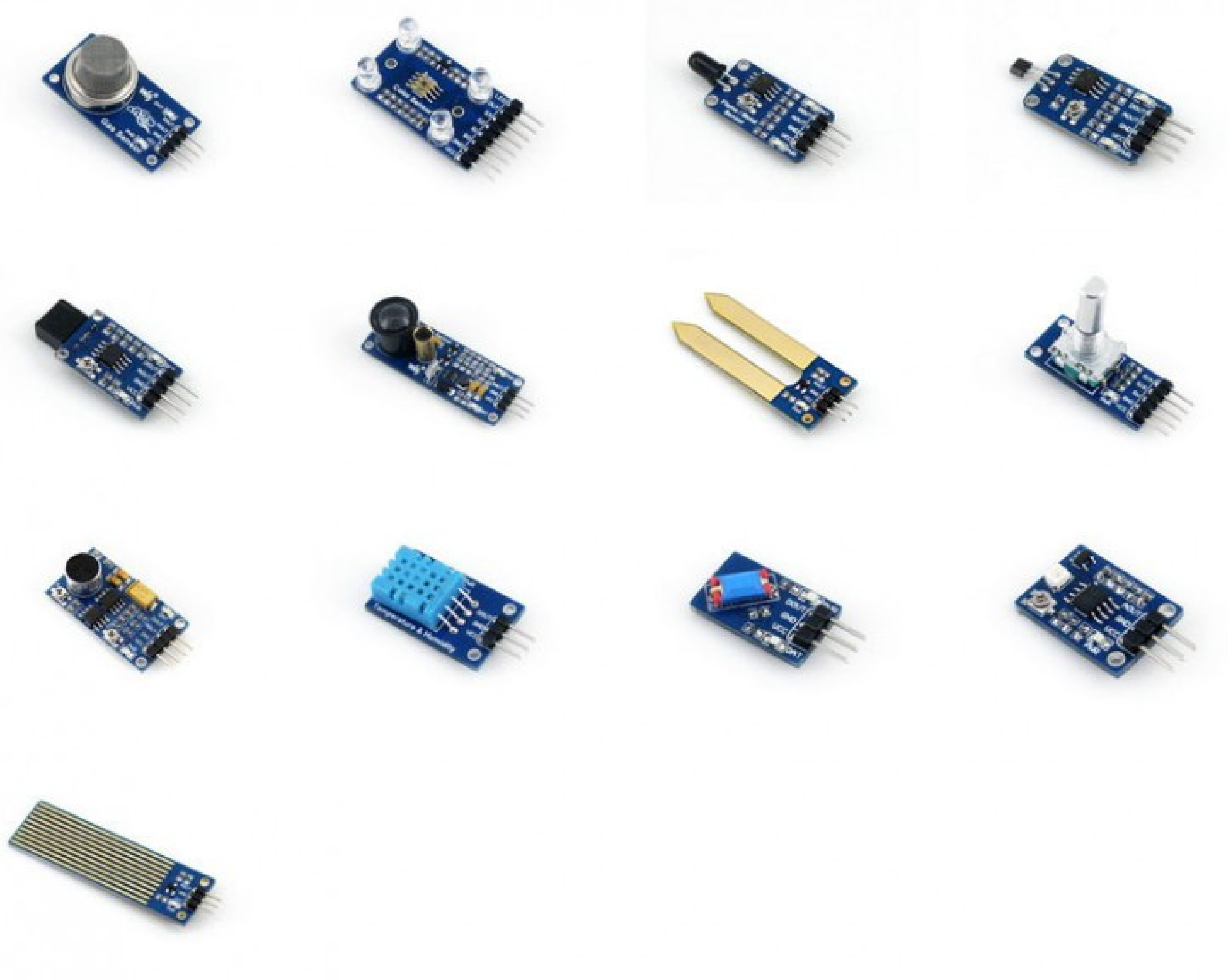
Webcams can be used for live chatting over the internet which is called **video conferencing**. Or live feed of anything happening anywhere.

MICROPHONE

Microphones are either **built into** the computer or are **external** devices connected through wires or wirelessly. It is used to input speech/sounds across various applications, for example to convert speech into text that can be used as commands in ways to interact with a computer, can be **used as a sensor** to pick up sound, or as ways to **communicate**. It is **faster** to read in text to type it, the voices you record could be altered, lets you multitask using your hands for something and your voice to perfom something else. But sound files could take a lot of space and it is **not always very accurate** at picking up sound.

SENSORS

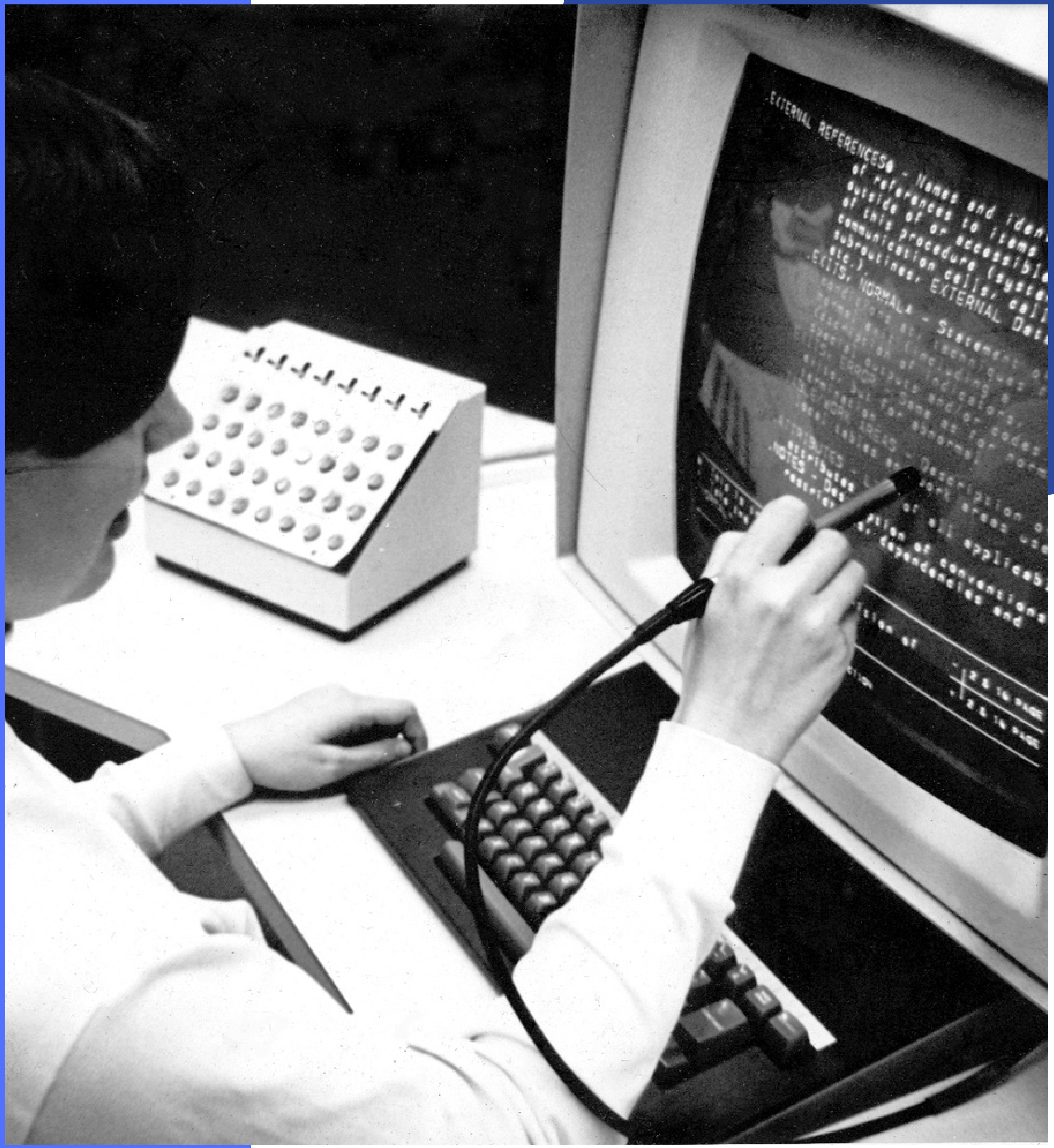
These are devices that input data to a computer, data being a measurement of physical quantity that is continuously changing, these data being **analogue** in nature they have to be **converted** to digital. These facilitate many things that human operators fail to do as accurate, it is continuous so it removes the need for human intervention, and they are automatic. But they are **fragile** and if damaged or dirty, they could **give false info**.



- Temperature
- Pressure
- Light
- Sound
- Humidity/moisture
- pH

LIGHT PENS

These are used with computers as an input, they were mostly used before the development of touch screens. They contain sensors that senses the progress of the pen on its surface and sends this info to the computer. It is rather a dated technology and it only works with **CRT monitors**, it is easy to use and it can be more accurate than touch, but it has lagging issues along with drawing inaccuracies.



2.2
DIRECT DATA ENTRY AND ASSOCIATED DEVICES

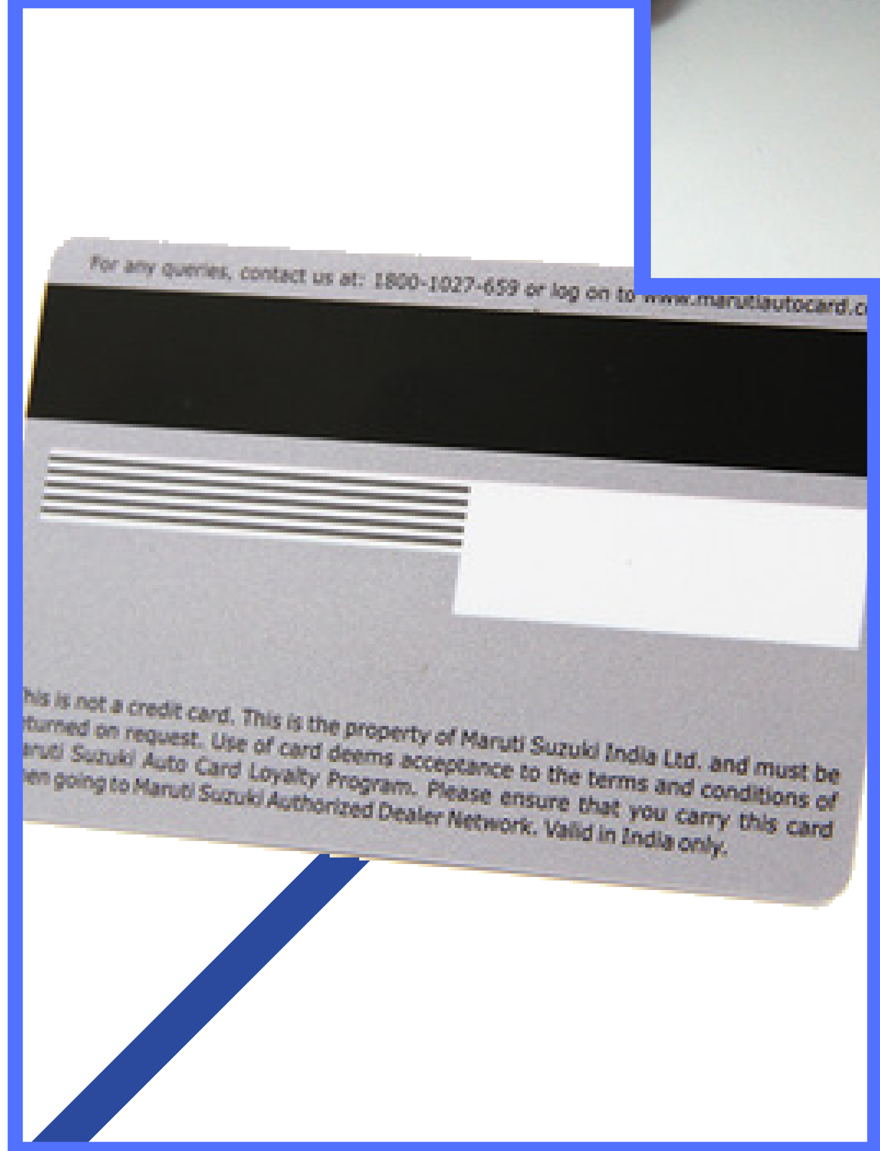
DIRECT DATA ENTRY (DDE) DEVICES ARE USED TO INPUT DATA INTO A COMPUTER WITHOUT THE NEED FOR VERY MUCH, IF ANY, HUMAN INTERACTION

MAGNETIC STRIPE READERS

These devices that read data stored on the **magnetic black lace** on the back of some cards. These cards can hold information such as bank info, hotel room key, library cards, etc..These devices read the data when swiped through the proper input place.

Advantages:

- Simple for people (no training needed)
- Inexpensive
- Fast data entry (just swipe)
- No error because no manual input of data
- Data can be altered on card easily



Disadvantages:

- Small storage capacity
- Magnetic fields can destroy its data
- Insecure- card writers and readers are easy to obtain
- It needs a reader to have its data used

CHIP & PIN READERS

A more secure type of cards that rely on a **chip** to store data, what makes it **more secure** is that the data on the chip is **encrypted**. The chip can also store and generate the details of the transaction each time they are used. When inserted to the reader, the user needs to use their **PIN** (Personal Identification Number) to use the card's data.

Advantages:

- Info is more secure than magnetic stripe because chips are harder to clone
- The chip holds more information than the magnetic stripe
- Chip and PIN readers can be wireless, so they can be brought to you

Disadvantages:

- 'Shoulder surfing' can cause people to know your PIN
- PIN can be forgotten

CONTACTLESS CARDS READERS

Contactless debit or credit cards allow customers to pay for items worth up to a certain amount of money without entering their PIN. All contactless cards have a small **chip** that emits **radio waves** embedded in them. The card is held within a few centimetres of the payment terminal to pay for an item.

Advantages:

- Faster transactions than magnetic stripes
- No data entry error (no PIN)
- The contactless card system uses 128-bit encryption
- The chip responds to the payment terminal reader with a unique number used for that transaction only

Disadvantages:

- More expensive than normal credit/debit cards
- A thief with a suitable reader could monitor your transactions
- Transactions are limited (e.g 50$ per transaction)

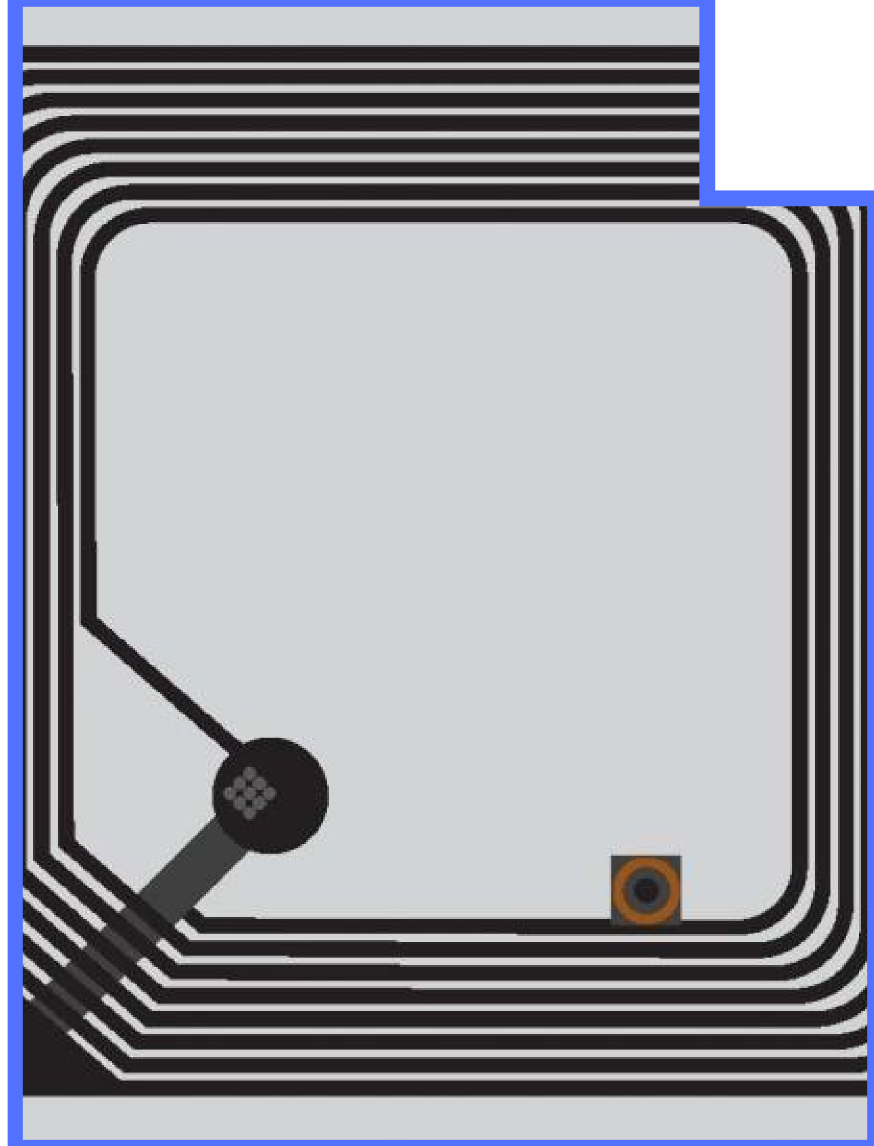
RFID READERS

RFID (Radio Frequency IDentification) reader will read data stored on RFID tags. Just like a barcode and magnetic stripe, an RFID device provides a **unique identifier** for the object it is attached to. Anything can have an RFID tag, shopping items, animals, train carriages, etc...

RFID has 3 elements:

- A scanning antenna
- A decoder to interpret data
- The tag.

The **tag** has an **antenna** and a **microchip** to hold the info. It is either a **passive** tag that uses the reader's **radio wave energy** to relay back the information.
Or it can be an **active** tag and be powered by a **battery** to relay back the information.



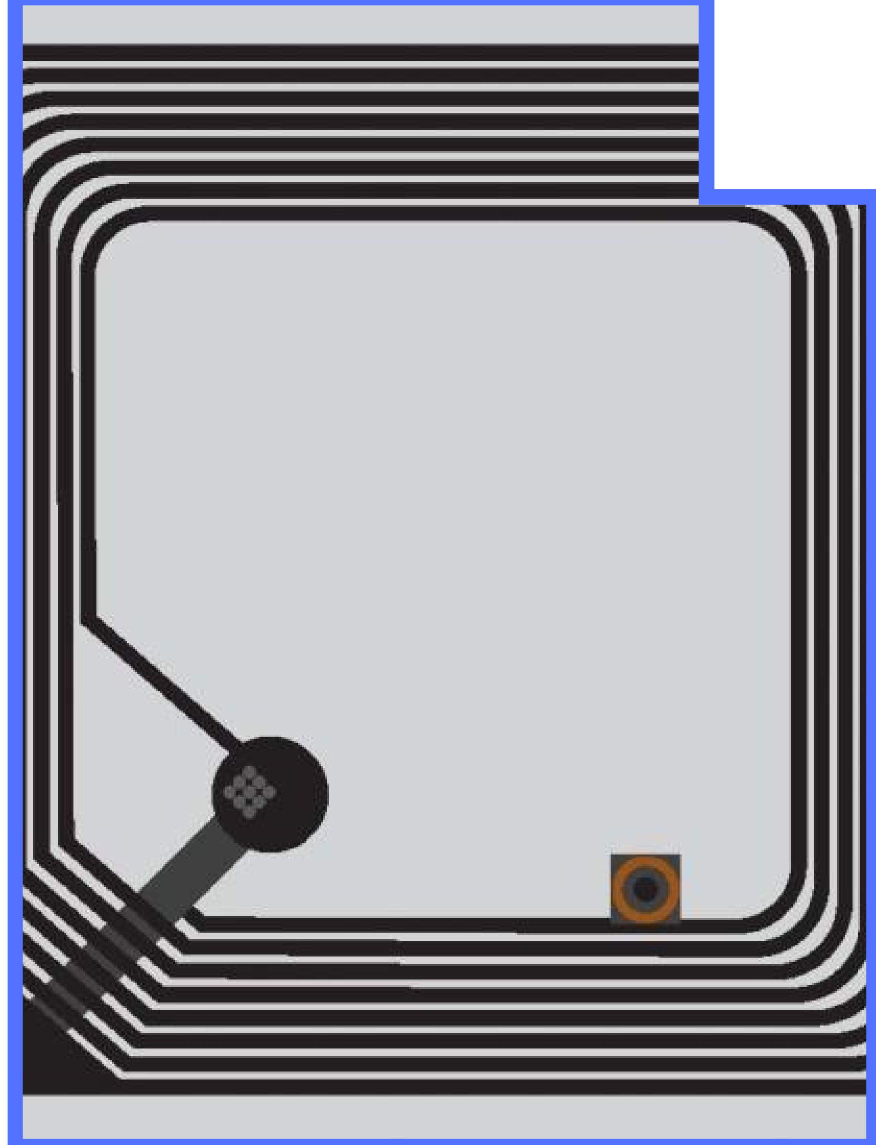
RFID READERS

Advantages:

- Tags do not need to be positioned precisely on a scanner
- They work from within a few meters of the scanner
- Many items can be scanned at the same time
- You can write data to RFID tags, not just read from them
- RFID tags can be used for tracking items

Disadvantages:

- Tags can be read without authentication of the owner, and can clone them
- Because RFID uses radio waves, they are relatively easy to jam or interrupt
- Tag collision (this is when the signals from two or more tags overlap, interfering with each other)



OCR

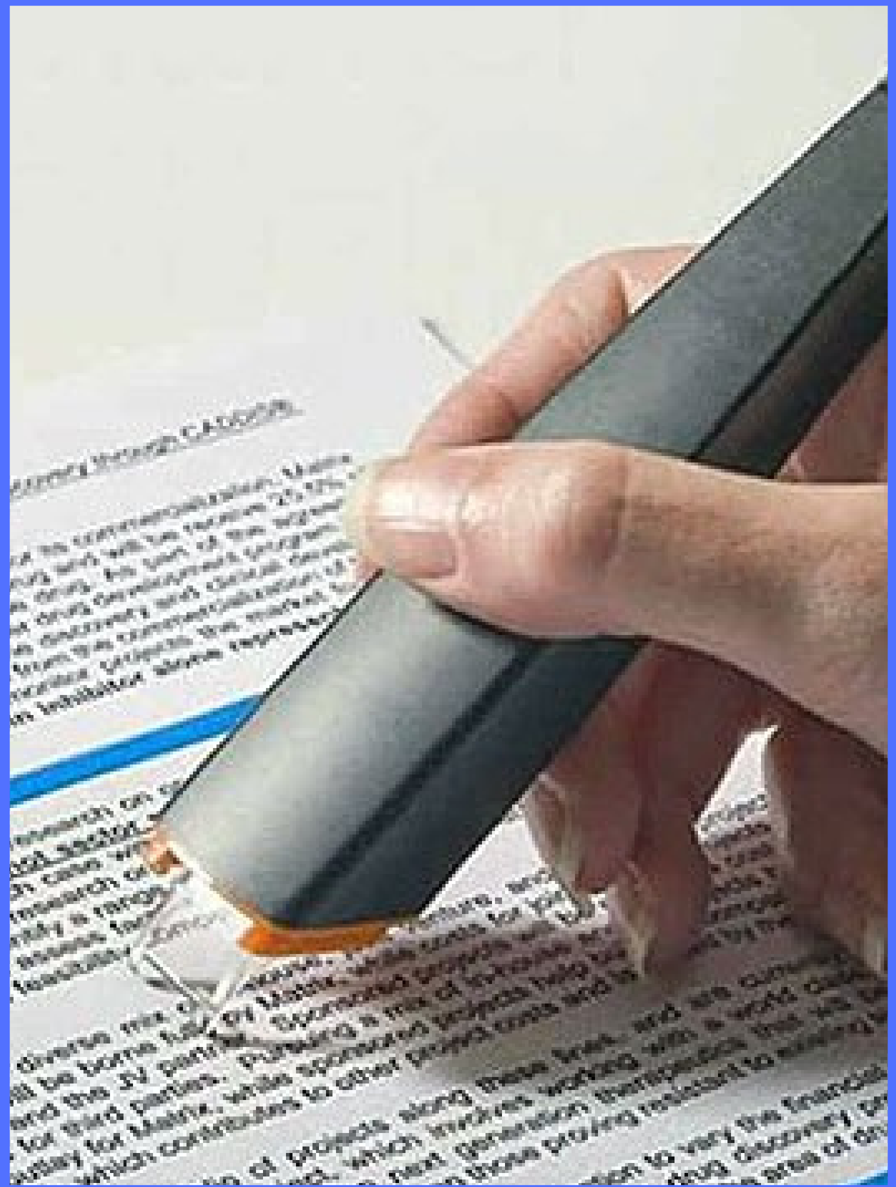
Optical Character Recognition is the name given to a device that **converts** the **text** on hard copy documents into an **electronic** form. OCR software converts this electronic data into a form that can then be used in various application packages, such as word processors or presentation software. One of the most recent uses is the processing of passports and identity cards to extract the text only and convert them into digital form.

Advantages:

- Fast way of entering hard-copy data
- It can avoid typing errors
- Cheaper than getting someone to input hard-copy data

Disadvantages:

- The hard-copy text can sometimes be hard to read by the device, so inaccuracy is a disadvantage



OMR

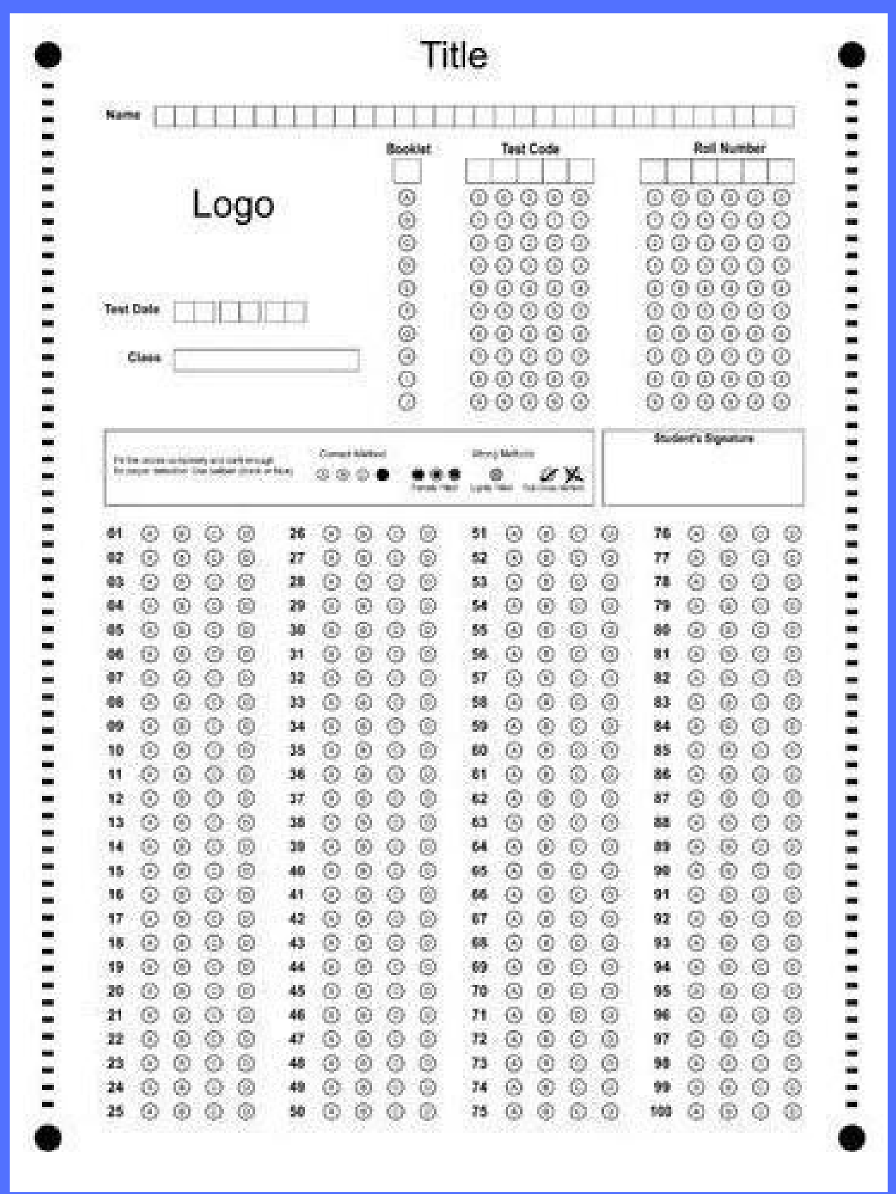
Optical mark recognition is a device which can **read marks** written in pen or pencil on a form.Used to read questionnaires, multiple-choice examination papers, voting papers and many other types of form where responses are registered in the form of lines or shaded areas. It is not interested in the shape of the marks, only **where they are**.

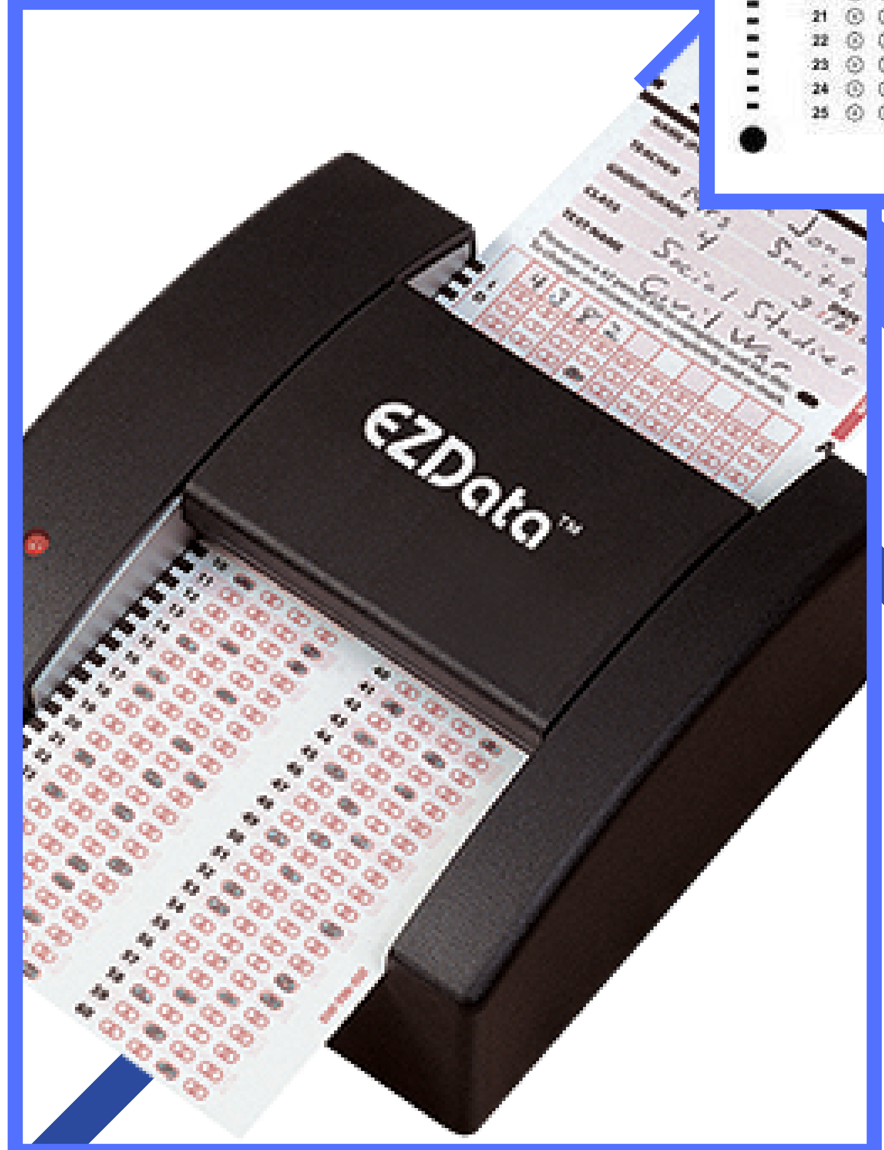
Advantages:

- Extremely fast to input data
- More accurate than humans checking papers

Disadvantages:

- There can be problems if they have not been filled in correctly
- Dirty marks on paper can cause inaccuracies
- Can only read marks, not text
- Because of the speed of the machines, paper jams can occur





BARCODE READERS

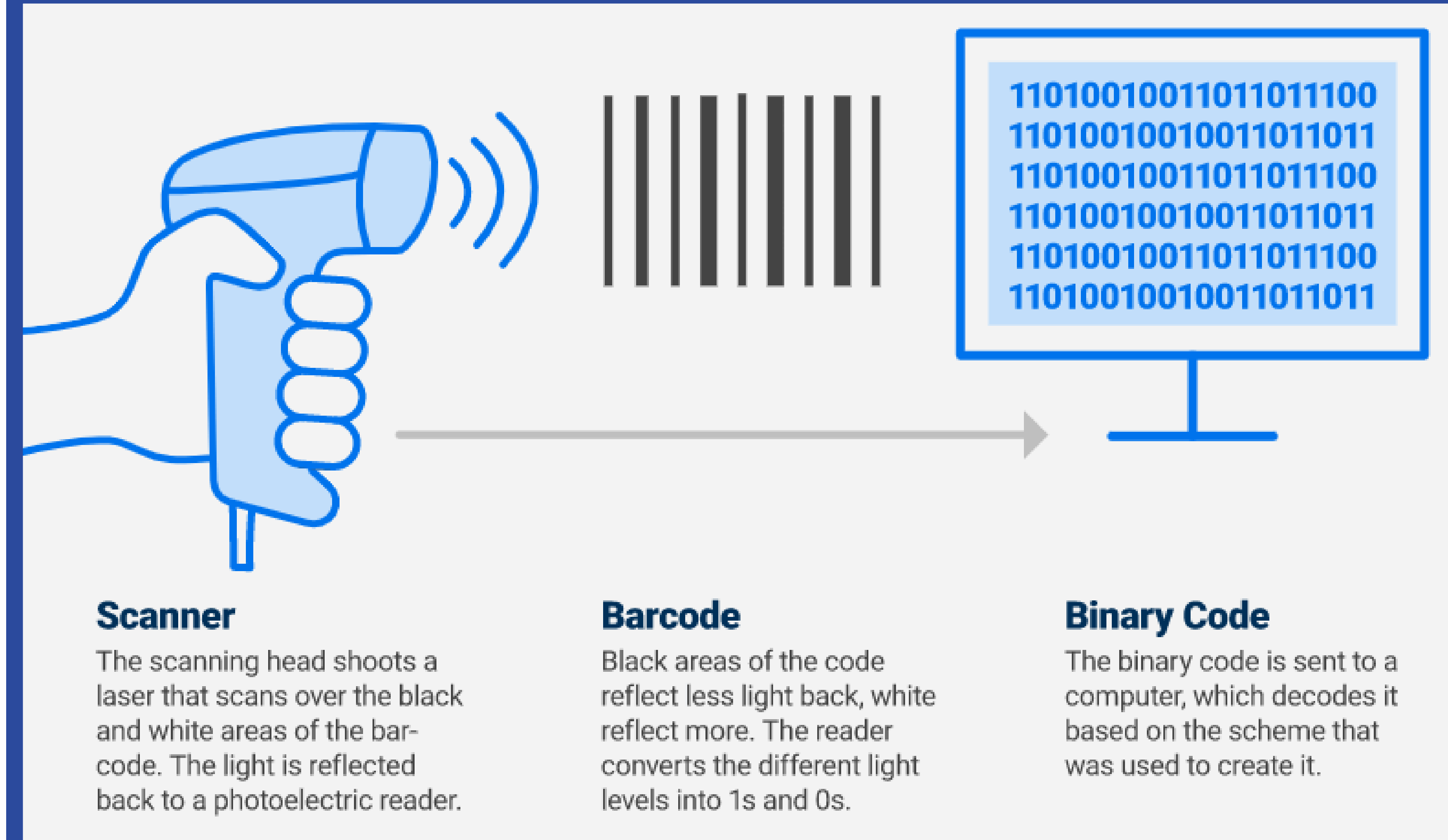
A barcode is a set of short parallel lines. The dark lines on a barcode are thick, medium, or thin. These **lines are read** by a scanner as a code number, which can **hold data** about the details of a certain product (**not price**). The readers are devices that shine a laser on the barcode and read the reflection of the laser to tell how thick the lines are. A typical barcode info includes: **Country of origin no, manufacturer no, item no, check digit**.

Advantages:

- Much faster than keying in data manually
- used as a way of recording safety testing of components
- They are a tried and trusted technology

Disadvantages:

- Not fool-proof (barcodes can be swapped around on items)
- Can be more easily damaged



QR SCANNERS

QR (**Quick Response**) codes are a type of two-dimensional bar code that can be read using a **camera** and **software** to interpret them and provide a link to the text, email, and website that they contain. A QR code consists of a block of small squares (light and dark) known as pixels. It can hold up to 7089 digits and allows **internet addresses** be encoded within them. This compares to the 30 digits which is the maximum for a barcode. The more data a QR code holds, the more **complex** they are.

Advantages:

- A lot of info can be stored without writing anything down
- Can be used for anything by anyone
- Can be scanned from any direction

Disadvantages:

- You need a smartphone with a camera and software
- Harmful code can be embedded within QR codes
- They are new and not everyone is aware of what they represent

CATCH-A RIDE!

Output devices and their uses

CRT Monitor

Cathode Ray Tubes are the least expensive type of monitor, they have been almost phased out anywhere due to being outdated, but they are talked about due to their compatibility with light pens. They are used in Computer Aided Design applications as light pens could be used on its screen to create designs, what also helps is that their screens are usually large to enable creation and modification of complex diagrams.

Advantages:

- Wider range of view angles than most LCD monitors
- Compatibility with light pens

Disadvantages:

- They are heavy and bulky
- As they get older they can cause fire if left unattended
- Higher power consumption
- Their screens can flicker, causing eyesight problems with use

LED and LCD Monitors

LED screens are made of tiny **Light Emitting Diodes** (LEDs). Each being assigned to a color, with its control of brightness a vast range of colors are produced. These are mostly used for large outdoor displays.
Many monitors and TV screens are advertised as LEDs, when in fact they are LCD screens backlit using LEDs.
LCD screens are the main output for most modern computers, and many of them offer touch screen input.

LCD screens are very power efficient, lightweight, they don't suffer from screen **image-burn** in nor flickering like in CRT monitors, and have **sharp image resolution**. Disadvantages to this type of screen is that it can be **inconsistent** in terms of color and contrast when viewing from various angles, motion blur, **lower contrast** than CRT, and can have problems with **uneven** lighting over the screen.

Touch screens (As output)

One of the few devices that can both be used as an **output** and an **input**, here we talk about it being an output device. Like smartphones, tablets, ATM at banks, or any other screen that displays information.
Faster entry of options than using a keyboard or a mouse, and makes choosing options very easy. It also is **user-friendly** for being too simple to use and you have an option to **expand** the display.
The cons of this device include the **limitations** in options, spread of **infections**, and **lack of fast response** if large amounts of data are being input or output.

Multi-media Projectors

These devices use signals, which are often received from a computer, or TV, and then the image from the source is **magnified** and projected onto a large screen. Many uses could be found for this but most commonly they are for presentations, advertisements, and home cinema systems.

This device provides use for avoiding to log into many computers to **view** something for **many people**, it is instead shown from a single source for everyone to see. What gives difficulty is that their images could be **fuzzy**, **costs**, and struggle to **set up**. Another disadvantage being that it relies on an **expensive fragile bulb** to show pictures.

Laser printer

These printers work by using a laser to draw the output on a drum and then a powdered ink (toner) is used to print the output. These have a high quality output, and they rely on **large buffer memories** to store document data before they are printed. They are more suitable in **offices** due to their **lack of noise** when operating, and if fast, high quality, and high volume printing is required then these are the best option.

Good traits of these printers include **fast printing**, well handling of **large print jobs**, consistent **high quality**, **cost effective** meaning that their **toner** cartridges last longer.

The traits that put this printer at a disadvantage is that they are only fast if several copies are being made so at **low volume** printing there **could be faster** printers, the color laser printers are more **expensive** to run and to buy, they can cause **health issues** by by producing ozone and volatile compounds due to their method of printing and the used toner type.

Inkjet printer

Inkjets **do not** have large buffer memories to store the document print job, so the printing is done **a bit at a time**, which causes the printing to sometimes pause, so it waits for the computer to send more data. At **low-output volumes** of printing it is advised to use inkjet, also when single paged high quality printing is needed.

Good thing about these printers other than their **high quality** output is that they are **cheaper** to buy, and **avoid producing volatile** organic compounds in comparison with the laser printers. They are also **lightweight** and **take less space**

However they are **not** very suitable for **high volume printing** as it tends to be slow due to their little buffer capacity, and they **run out** of ink cartridges **too fast** which causes another disadvantage which is ink costs. It also sometimes causes **smudging** when printing.

Dot matrix printer

They are a type of **impact printer** where a print head of matrix pins is pressed against an inked ribbon. These tend to be slow and noisy, their output is also not that good compared to the other two. They can work under **dirty and noisy conditions** where quality is not an essence.

Unlike inkjet and laser printers, they **can** be used in **rough conditions**, with them **carbon copies** could be produced, they are **cost effective** in terms of maintenance, and **easy to use** when needed for long print jobs like wages slips.

The cons of this machinery include its **noisiness**, **initial costs are more**, and it is **very slow** with offering **low quality** printing.

3D printers

3D printers are devices that produce solid objects designed using **CAD software**. The solid object is built up layer by layer using materials such as powdered resin, powdered metal, paper or ceramic. 3D printing is regarded as being possibly the next 'industrial revolution' because it will change the manufacturing methods in many industries. Examples being prosthetic limbs, Fashion and art, aerospace and parts no longer in production.

Advantages start from its **efficiency** and **quickness** in manufacturing any item, **less cost** when considering the labor costs and other costs involved in manufacturing a product, **Medical aspect benefits**, and its use to enable producing **parts** for items that are **no longer being produced**.

Disadvantages come from misuse of this technology, either **infringing copyright** or being used for manufacturing **dangerous and illegal items** because the limitations with 3D printers are minimal, and finally there is the potential for **job losses** if this technology takes over from some types of manufacturing.

Plotters

A graph plotter although prints on paper, it works differently to printers, instead of ink or toners they use **pen**, **pencil** or **marker** to draw multiple continuous lines rather than a series of lines. The paper size here varies from an A4 to several meters long.

Uses of this device include architectural and engineering drawings, and also cartoon/ animation characters. They have a very **high quality** output by doing so, they also **very accurate** at large monochrome and color drawings, and not just on paper but also **cardboards, steel, wood**...

These are although being phased out due to there being wide-format inkjet printers at lower costs. The downsides are that they are **very slow**, and take up **a lot of space** compared to a printer, and they are initially **costly** but after purchase they are cheap to run.

Speakers

Output devices that **produce sound**. They are connected to a computer system, and digitised sound stored on a file is converted when data flows through a **DAC** where it is changed to an electric current then to an amplifier, in the end fed to a loudspeaker where it becomes sound.

A very simple technology found on all phones and built into most computers, projectors. They amplify sound louder than the original, which can help in large audience places, also visually impaired people.

Could be **misused** in the hands of annoying and irresponsible people. High quality sound demands **high costs,** and another thing can be that they could take **a lot of desk space**.

Actuators

An actuator is a **mechanical** or electromechanical device that is used by a computer to **control another device**, like a conveyer belt, valve or a motor. Uses of actuators include controlling of motors, pumps switches, LED, buzzers and so on, so they allow computers to control physical devices.

They are **very cheap** and enable **remote operation**. But they have a few disadvantages being an additional device in the system that **can go wrong**, and due to them being analogue devices, the computer signals **need to be** converted by a **DAC**.

Fin